# Vector-Valued Functions

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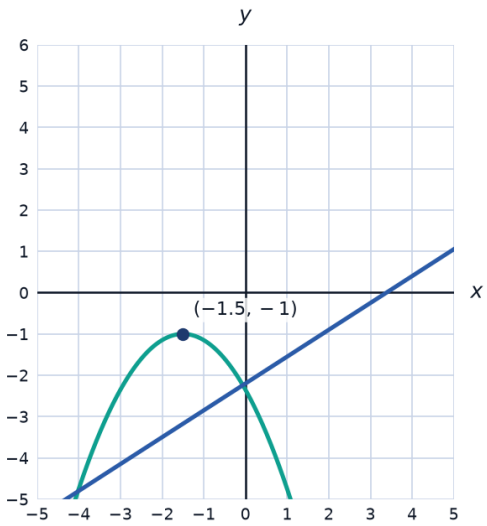
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Position, velocity, and acceleration vectors

A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

**2** [Foundational] • Target: Position, velocity, and acceleration vectors

For the same particle, find the speed at $t = 2$.




**3** [Foundational] • Target: Integrating vector-valued functions

Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

**4** [Foundational] • Target: Total distance traveled

A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.

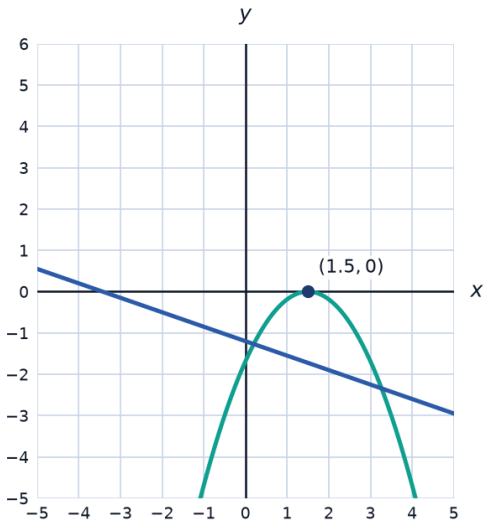
**5** [Foundational] • Target: Vectors and parametric motion combined

A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

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**6** [Foundational] • Target: Vectors and parametric motion combined

For the particle above, find the magnitude of the acceleration vector at $t = 0$.




**7** [Foundational] • Target: Unit tangent vectors

For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

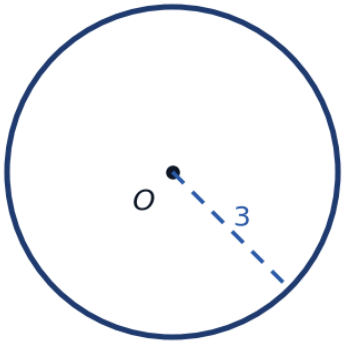
## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Unit tangent vectors

Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude $1$.

**9** [Intermediate] • Target: Motion on a circle and centripetal acceleration

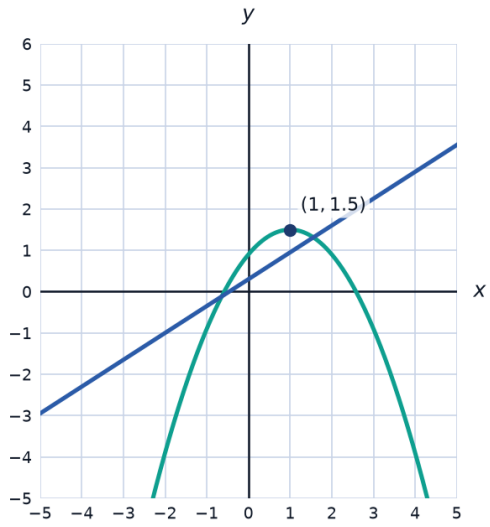
A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).




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**10** [Intermediate] • Target: Motion on a circle and centripetal acceleration

For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all $t$, and interpret this geometrically.




**11** [Intermediate] • Target: Displacement vs. distance for vector motion

A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

**12** [Intermediate] • Target: Displacement vs. distance for vector motion

Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

**13** [Intermediate] • Target: Vector operations review

Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

**14** [Intermediate] • Target: Curvature-related speed problems

A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times $t$ when the particle's speed is exactly $\sqrt{13}$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

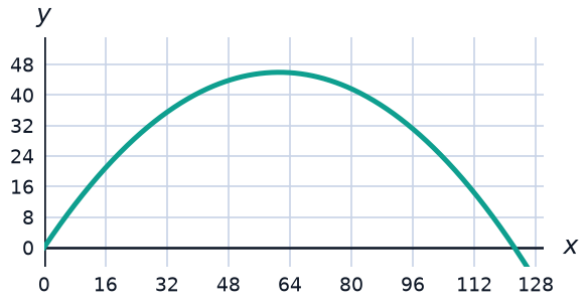
**15** [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

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**16** [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A projectile is launched with position $\vec{r}(t) = \langle 20t,\ 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is $0$).




**17** [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 1.01 Integration by Parts

For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

**18** [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A particle moves along $\vec{r}(t) = \langle t^3 - 3t,\ t^2 - 4 \rangle$. Find all times $t$ when the particle is momentarily at rest (both velocity components equal $0$ simultaneously).

**19** [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

For the same particle, find the times when only the horizontal velocity component is $0$.

**20** [Challenging] • Target: Vector-valued functions and arc length • Cross-Lesson: 3.01 Parametric Equations

State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Position, velocity, and acceleration vectors

A particle moves along $\vec{r}(t) = \langle 5t^2,\ 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

**a.** 26

**b.** 28

**c.** 34

**d.** 13

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**22** [Foundational] • Target: Integrating vector-valued functions

A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 + 3 \rangle$. Find the positive time $t$ when the horizontal velocity component is $0$.

**a.** $-2$

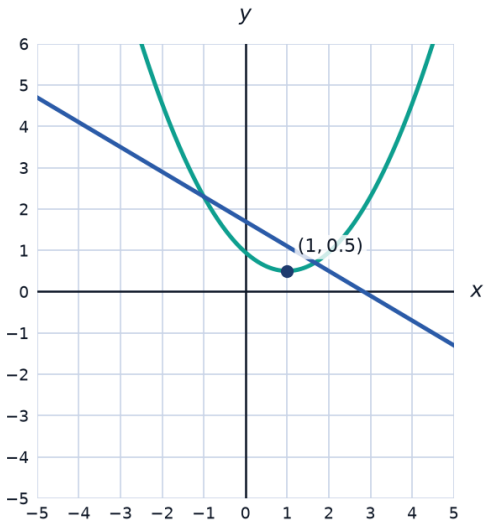
**b.** $6$

**c.** $2$

**d.** $3$

**23** [Foundational] • Target: Total distance traveled

A particle moves along $\vec{r}(t) = \langle 7t, 48t - 4t^2 \rangle$. Find the time $t$ when the vertical velocity component is $0$ (the time of maximum height).




**a.** $7$

**b.** $6$

**c.** $12$

**d.** $5$

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**24** [Foundational] • Target: Vectors and parametric motion combined

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of $t$ at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

**a.** Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$

**b.** Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

**c.** Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

**d.** Rest at $t = \pi$, acceleration $a = 0$

**25** [Foundational] • Target: Unit tangent vectors

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

**a.** $P(x)$ is always positive and increasing.

**b.** $P(x)$ is always negative and decreasing.

**c.** $P(x)$ is always positive and decreasing.

**d.** $P(x)$ is always negative and increasing.

**26** [Foundational] • Target: Motion on a circle and centripetal acceleration

Unit tangent vector $\mathbf{T}(t) = \mathbf{r}'(t)/|\mathbf{r}'(t)|$. It has magnitude:

**a.** 1

**b.** 0

**c.** $|r'|$

**d.** $t$

**27** [Foundational] • Target: Displacement vs. distance for vector motion

For a circle of radius $R$, curvature $\kappa =$

**a.** $R$

**b.** $2R$

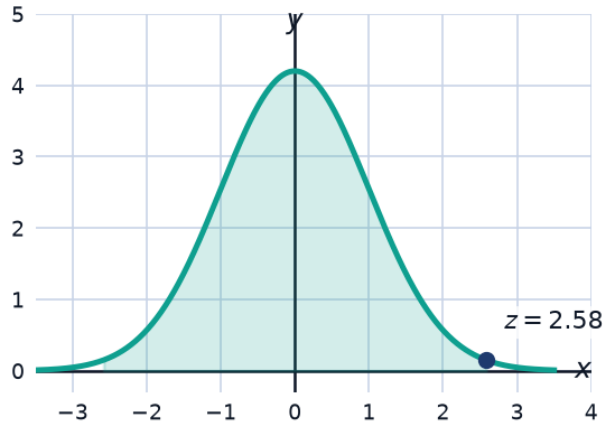
**c.** $1/R^2$

**d.** $1/R$

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**28** [Foundational] • Target: Vector operations review

Tangent plane at $(0, 0, 1)$ on unit sphere: $z = 1$. Normal direction:




**a.** $(0, 1, 0)$

**b.** $(1, 0, 0)$

**c.** $(0, 0, 1)$

**d.** $(1, 1, 1)$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Curvature-related speed problems

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where $t$ is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

**a.** $t = 1$ and $t = 3$

**b.** $t = 1$ and $t = 2$

**c.** $t = 2$ and $t = 3$

**d.** $t = 1, t = 2,$ and $t = 3$

**30** [Intermediate] • Target: Projectile motion as a vector-valued function

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

**a.** $-4$

**b.** $4$

**c.** $8$

**d.** $-8$

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**31** [Intermediate] • Target: Finding when a particle is at rest

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?

**a.** $-1$ m/s$^2$

**b.** $0$ m/s$^2$

**c.** $1$ m/s$^2$

**d.** $-2$ m/s$^2$

**32** [Intermediate] • Target: Vector-valued functions and arc length

A particle's velocity is given by the function $v(t) = 5$ for $0 \le t \le 4$. Find the total displacement of the particle over this time interval.

**a.** 10

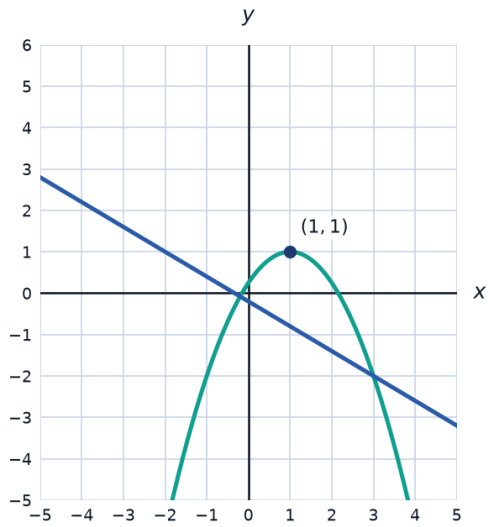
**b.** 15

**c.** 20

**d.** 25

**33** [Intermediate] • Target: Position, velocity, and acceleration vectors

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:




**a.** $|\mathbf{b} - \mathbf{a}|$

**b.** $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$

**c.** $|\mathbf{d}_1 - \mathbf{d}_2|$

**d.** $0$

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**34** [Intermediate] • Target: Integrating vector-valued functions

Parametric surface $\mathbf{r}(u, v) = (u\cos v, u\sin v, u^2)$. At $(u, v) = (1, 0)$:

**a.** tangent plane is $z = x^2$

**b.** tangent plane is $z = 2x - 1$

**c.** normal is $(0, 0, 1)$

**d.** surface is a sphere

**35** [Intermediate] • Target: Total distance traveled

Find the curvature of $\mathbf{r}(t) = (t, t^2, 0)$ at $t = 0$:

**a.** 2

**b.** 1

**c.** 0

**d.** 1/2

**36** [Intermediate] • Target: Vectors and parametric motion combined

For $\mathbf{r}(t) = \langle 2t, 3t^2, t\rangle$, find the normal component of acceleration at $t = 1$.

**a.** $\frac{6\sqrt{205}}{41}$

**b.** $\frac{36\sqrt{41}}{41}$

**c.** 6

**d.** 36

**37** [Intermediate] • Target: Unit tangent vectors

For $\mathbf{r}(t) = \langle 2t, t^2, t\rangle$, find the tangential component of acceleration at $t = 2$.

**a.** $\frac{2\sqrt{105}}{21}$

**b.** $\frac{8\sqrt{21}}{21}$

**c.** 2

**d.** 8

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Motion on a circle and centripetal acceleration • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The torsion $\tau$ of a space curve measures how fast the curve twists out of its osculating plane. For a helix $\mathbf{r} = (a\cos t, a\sin t, bt)$, as $b \to 0$ (curve flattens to a circle), $\tau \to$

**a.** $1/a$

**b.** 0

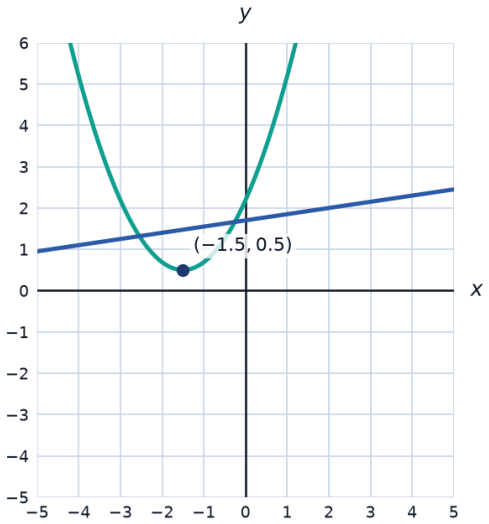
**c.** $\infty$

**d.** $b$

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**39** [Challenging] • Target: Displacement vs. distance for vector motion • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Distance from point $P = (1, 2, 3)$ to plane $2x + y - 2z = 5$:




**a.** $\frac{7}{3}$

**b.** $\frac{2}{3}$

**c.** $1$

**d.** $3$

**40** [Challenging] • Target: Vector operations review • Cross-Lesson: 3.01 Parametric Equations

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

**a.** $3$

**b.** $3V = 4\pi$

**c.** $4\pi$

**d.** $12\pi$

**41** [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 4.01 Polar Curves and Derivatives

Position vector of midpoint of segment from $A(1, 2, 3)$ to $B(5, 6, 7)$:

**a.** $(6, 8, 10)$

**b.** $(2, 2, 2)$

**c.** $(3, 4, 5)$

**d.** $(4, 4, 4)$

**42** [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 4.02 Area in Polar Coordinates

Curvature of helix $\mathbf{r} = (a\cos t, a\sin t, bt)$:

**a.** $ab$

**b.** $b/(a^2 + b^2)$

**c.** $1/a$

**d.** $a/(a^2 + b^2)$

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**43** [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Potential function for $\mathbf{F} = (2xy, x^2 + 3y^2, 0)$:

**a.** $f = x^2y + y^3$

**b.** $f = x^2y^2$

**c.** $f = xy^2$

**d.** $f = x^2 + y^2$

**44** [Challenging] • Target: Vector-valued functions and arc length • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Find the distance from $(2,\ -2,\ 0)$ to the plane $-3y + z = -4$.

**a.** $10$

**b.** $\frac{\sqrt{10}}{5}$

**c.** $1$

**d.** $\sqrt{10}$

**45** [Challenging] • Target: Position, velocity, and acceleration vectors • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Stokes' theorem for $\mathbf{F} = (y,\ -x, 0)$ over unit disk $z = 0$: $\oint \mathbf{F} \cdot d\mathbf{r} =$

**a.** $0$

**b.** $2\pi$

**c.** $-2\pi$

**d.** $\pi$

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